

Received 7 May 2019; revised 8 July 2019; accepted 13 August 2019. Date of publication 22 August 2019; date of current version 30 September 2019.
The review of this article was arranged by Editor K. Shenai.

Digital Object Identifier 10.1109/JEDS.2019.2937008

Stable Operation of AlGaIn/GaN HEMTs for 25 h at 400°C in air

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This work was supported in part by the Knut and Alice Wallenberg Foundation, in part by the National Science Foundation Engineering Research Center for Power Optimization of Electro-Thermal Systems (POETS) under Award EEC-1449548, in part by the National Nanotechnology Coordinated Infrastructure (NNCI) under Award ECCS-1542152, and in part by the National Aeronautics and Space Administration under Award NNX17AG44G and Award NNX17AG42G.

ABSTRACT Extreme environments such as the Venus atmosphere are among the emerging applications that demand electronics that can withstand high-temperature oxidizing conditions. While wide-bandgap technologies for integrated electronics have been developed so far, they either suffer from gate oxide and threshold voltage (V_{th}) degradation over temperature, large power supply requirements, or intrinsic base current. In this letter, AlGaIn/GaN high electron mobility transistors (HEMTs) are suggested as an alternative platform for integrated sensors and analog circuits in extreme environments in oxidizing air atmosphere over a wide temperature range from 22°C to 400°C. An optimal biasing region, with a peak of transconductance ($g_{m,peak}$) at -2.3 V with a negligible shift over the temperature range was observed. Moreover, remarkably low V_{th} variation of 0.9% was observed, enabling the design of analog circuits that can operate over the entire temperature range. Finally, the operation of the devices at 400°C and 500°C over 25 hours was experimentally studied, demonstrating the stability of the DC characteristics after the 5 hours of burn-in, at 400°C.

INDEX TERMS Gallium nitride (GaN), high electron mobility transistor (HEMT), high-temperature electronics.

I. INTRODUCTION

Wide-bandgap semiconductors such as silicon carbide (SiC) and gallium nitride (GaN) have proven to be viable candidates for operation in extreme environments due to their superior electronic properties, namely low intrinsic carrier concentration [1], [2]. Wide-bandgap electronics have recently found use in aviation, space exploration, automotive and deep-well drilling applications. NASA Glenn Research center (GRC) has studied SiC JFETs (junction field effect transistors) for harsh environments for approximately a decade [3] and demonstrated reliable electronics operational at 500°C for one year [4]. High-temperature devices and circuits in SiC have also been studied in n-type metal-oxide semiconductor (NMOS) and complementary metal-oxide

semiconductor (CMOS) [5]–[8], and BJT (bipolar junction transistor) [9]–[18]. The high-temperature SiC MOSFETs suffer from gate oxide and V_{th} degradation, while SiC JFETs demand large power supplies and SiC BJTs need a constant base current to operate. To address the need for low-power and high-temperature electronics, AlGaIn/GaN high electron mobility transistor (HEMTs) are suggested.

AlGaIn/GaN HEMTs are a promising candidate for implementation of integrated electronics [19]–[22], and a few studies have discussed the high-temperature capability of the HEMTs [23]–[25]. However, as compared with SiC, AlGaIn/GaN HEMTs have not been sufficiently matured for implementation of integrated circuits (ICs) for extreme temperatures. Previous work has investigated the

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